TEMASEK POLYTECHNIC

SCHOOL OF INFORMATICS & IT

Diploma in Applied Artificial Intelligence

Cloud Technologies for AI (CAI2C09)

AY2026/2027 April Semester

Project Specifications

Introduction

The purpose of this project is to explore how **cloud services** can be used to **build practical solutions for real-world problem statements**. You will use **AWS** as the designated cloud platform to design, build, and deploy a functional web application.

By the end of this project, you are expected to:

- Design the architecture of a web-based solution
- Assemble and implement the solution using relevant cloud services
- Evaluate its performance and functionality
- Deploy the final product on AWS

The project comprises the following components:

1. Project Proposal (Formative, not graded)
2. Project Deliverables (70% of subject final grade)
   a. Application (40%)
   b. Report (20%)
   c. Presentation & Demonstration (10%)

Note that this is an **individual** project.

This project carries a weightage of 70% of the subject's final grade.

Project Scope

You are required to propose, design, build, and test an **image recognition web application** that will be deployed on the Amazon Web Services (AWS) cloud platform.

There is no restriction on the nature of the web application. You are encouraged to select a use case that **aligns with your personal interests or addresses a real-world challenge**. Examples include:

- A plant disease recognition tool that uses images of leaves to identify potential infections and stores results in a cloud database.
- A student attendance tracker that scans photographs of identification cards and logs attendance records automatically.
- A vehicle damage assessment portal where users upload images of car damage for AI-based assessment.
- A tourism site recommender that suggests destinations based on uploaded scenic photographs.

You are free to propose any application within the constraints of AWS that uses image recognition meaningfully. You are supposed to aim for originality, practical value and effective use of AWS cloud services.

Your solution should demonstrate the following capabilities:

1. **Implementation of cloud security**.
   - Should include but is not limited to secure handling of data, proper access rights, proper user authentication

2. **Full-stack web development**
   - Visually appealing and user-friendly interface
   - Dynamic display and data entry features
   - Proper integration with a backend database

3. **Serverless deployment**
   - Your website should be hosted on AWS S3

4. **Cloud database integration**
   - Use of Amazon RDS (e.g., MySQL) or DynamoDB for backend storage

5. **Use of AWS Rekognition**
   - Integration of AWS Rekognition for image analysis and recognition tasks

6. **Integration of Additional AWS Services**
   - Incorporation of additional AWS services into your website. You will be required to justify in your project report and presentation how the chosen AWS services add value to your solution in terms of interactivity and functionality.
   - The following services are not considered as additional cloud services:
     a. S3
     b. RDS-MySQL
     c. Lambda
     d. API Gateway
     e. AWS Rekognition

You are required to use AWS exclusively for this project. External Generative AI APIs are optional and will not replace required AWS services in grading. Assessment will be based solely on the features and services available within AWS.

Feel free to consult the **AWS documentation** for the latest information regarding the learner environment, including **region restrictions** and **availability of services**. Please note that this information is for reference only, as **certain services listed as available may not be accessible** through your account. **Your tutor remains the final authority for confirmation** regarding service availability.

Project Proposal (Formative, Not graded)

You are required to propose a real-world problem scenario that your website will address. The website will be developed and deployed on the AWS cloud platform.

Although this proposal is not graded, it must be shared and discussed with your tutor. Your tutor will review the selected topic, provide feedback, and advise whether you should proceed with the proposed website development.

Once submitted via TPLMS, your proposal will serve as the official basis for your project, and the specifications will be considered locked in. You are **encouraged to stay aligned with the main theme outlined in your proposal**. While **minor deviations are acceptable**, significant changes to the project scope should be avoided.

Your proposal should **clearly articulate the problem statement**, highlighting the specific challenge or opportunity that your project intends to address. For instance, the proposal may describe a need to automate the sorting of digital images based on recognised objects, which could significantly enhance media storage efficiency.

The proposed solution must outline the intended approach, including **a breakdown of the technologies and AWS services** that will be used. To ensure clarity and completeness, your proposal should address the **4W1H** framework:

- **Who** will use the website?
  (e.g., professional photographers, media archivists)
- **What** features will the website offer?
  (e.g., automatic image tagging and categorisation)
- **When** will key components be developed and implemented?
  (e.g., a realistic project timeline or work plan)
- **Where** will data be stored and processed?
  (e.g., AWS S3 for storage, AWS Rekognition for analysis)
- **How** will the solution function technically?
  (e.g., users upload images to S3, which triggers analysis via Lambda and stores results in RDS)

To help you craft a strong problem statement, consider referring to the following resources:

- Writing a Problem Statement: 10 Effective Tips
- How to Write an Effective Problem Statement – iSixSigma

The **cloud services proposed** must be **relevant and meaningful** to your solution (e.g. AWS Rekognition for image analysis). If you are unsure about the suitability of a particular service, please consult your tutor for guidance.

To ensure that your tutor gains a clear understanding of your chosen topic and project workplan, your proposal must be written in a **professional and coherent manner**, with careful attention to **language, grammar, formatting**, and the **logical organisation of content**. The use of **visual aids**, such as **diagrams or flowcharts**, is encouraged to improve clarity and enhance the reader's comprehension of your proposed solution.

You are strongly encouraged to make use of the **proposal template** available on **TPLMS** (under the *Project* section) to guide the structure and presentation of your submission.

The **final version of the proposal**, as agreed upon between you and your tutor, must be **uploaded to TPLMS** (under the *Project* section) by **Friday, 5th June 2026, 1700 hours**. The proposal should be submitted in **Microsoft Word (.docx)** format.

Project Deliverables – Application (40%)

For this component, you are to develop the website based on your project proposal. Websites that are done through minor modifications of the lab codes taught in the lectures will not be accepted. You are expected to exercise due diligence in applying the knowledge gained from lessons to develop an original web application.

Your website must consist of the following implementations:

1. Serverless deployment to S3
2. Utilisation of cloud database (e.g., RDS-MySQL)
3. At least 2 of the database operations for CRUD (at least 1 SELECT operation and 1 INSERT operation)
4. Basic cloud security (e.g. data protection)
5. Have an image recognition component using AWS Rekognition that is related to the nature of the website

You are also encouraged to expand your learning by exploring the use of more cloud service(s) that can add value and interactivity to your website.

There are 2 submissions for this component:

1. Deliverables Part 1
2. Deliverables Part 2

Submission – Deliverables Part 1(10%)

For this submission, you are required to submit the complete source code for your website before incorporating any AWS services. You are to include **JavaScript functions** that are required to call AWS API URLs (e.g., CRUD APIs, image recognition API). You are not required to include working URLs for this deliverable. Static hardcoded response simulating API results should be used. If you are uncertain, your tutor should be consulted for guidance. Your website should also have an appealing visual design.

You are to **zip up your website files (e.g, HTML, CSS and JavaScript files)** and **submit to TPLMS** by **Friday 17th July 2026, 1700 hours** under the *Project* section.

Important: Your zip file **MUST** be named using the following format:

<matric number>_<name>.zip

Example:

2300000A_TanChuChu.zip

Please ensure you have a **backup copy** of your zip file in case there is a problem with the online submission.

Submission – Deliverables Part 2 (30%)

For this submission, your website should be **deployed to S3** and **fully functional with all proposed AWS services** integrated.

You are to submit your complete website codes to TPLMS by **Friday 14th August 2026, 1700 hours** under the *Project* section. Your submission should include your **entire web application codes**.

For Lambda functions, copy each function and paste them into separate .txt files.

Important: Your zip file **MUST** be named using the following format:

<matric number>_<name>.zip

Example:

2300000A_TanChuChu.zip

Please ensure you have a **backup copy** of your zip file in case there is a problem with the online submission.

Refer to the **marking rubrics below** for the marking criteria.

Project Deliverables – Report (20%)

After deploying your website, you are required to write and submit a report documenting the development and deployment of your website to AWS.

Your report should include the following sections:

a. Introduction of website
   - Brief overview of website
   - Screenshots of all the pages of the website

b. Development process
   - Documentation of the development process on AWS Rekognition

c. Deployment process
   - Outline of each step necessary for setting up the website on S3
   - Include configuration settings and service dependencies

d. Setup guide for AWS services
   - Setup / configuration guide of the services
   - Explain how each service is configured and how they interact with one another

If you have implemented additional cloud services, you are to include the development and deployment process of the services. Do include visual aids to support the content of your report.

Your report should be **submitted to TPLMS** by **Monday, 17th August 2026, 0900 hours** under the *Project* section. Your report should be in **Microsoft Word (.docx)** format. **Do not** submit your report in PDF format.

You can use the report template (available on TPLMS, under *Project* section) to help you get started.

Important: Your zip file **MUST** be named using the following format:

<matric number>_<name>.docx

Example:

2300000A_TanChuChu.docx

Please ensure you have a **backup copy** of your report in case there is a problem with the online submission.

Refer to the **marking rubrics below** for the marking criteria.

Project Presentation & Demonstration (10%)

Demonstrate and present your deliverable with appropriate presentation slides on **Week 18 (Week starting 17th August 2026)**. Keep your presentation and demonstration to 10 minutes.

Your presentation and demonstration should include the development process and explanation of the cloud services used.

You are required to present your presentation using a **PowerPoint (.pptx)** file, not exceeding 15 slides (excluding cover slide, references, appendices). You can use the presentation template (available on TPLMS, under Assessment section) to help you get started. You are reminded to test all the services for the final presentation and demonstration.

You are also required to do a **video recording** of your demonstration, of **no more than 5 minutes**. In the video, you are to explain the services that you have utilised and how they are being used. It is highly recommended that you use MS Teams for the recording, using the Presenter mode so that your face is captured in the video. This video will serve as a backup should the cloud service not be available during the presentation. The video should be uploaded to YouTube and set to "unlisted". You are expected to include the link to the video in your presentation slides.

Submit your presentation slides with your Project Report to TPLMS by **Monday, 17th August 2026, 0900 hours** under Project section.

Refer to the **marking rubrics below** for the marking criteria.

Penalty for Late Submission

Late and <1 day: 10% deduction from absolute mark given for the project

Late >=1 and <2 days : 20% deduction from absolute mark

Late >=2 days : No marks awarded

Note that "day" includes non-working days (Saturdays, Sundays, and Public Holidays).

General MC/LOA is NOT considered as valid reason for extended deadline of submission.

Academic-Related Offences

Type of Offence	Recommended Punitive actions
• **Cheating on Semestral Examination** (including possession of unauthorised materials as verified by the Examiner as relevant to the examination paper) • **Cheating on Coursework** such as projects, term tests and assessments • **Plagiarism** (refer to Plagiarism Policy for the definition, examples and ways to avoid plagiarism)	Punitive actions could include the following: • Warning letter • Failing the assessment item/subject • Suspension from course of study for a minimum period of one semester • Dismissal from the Polytechnic

Marking Criteria – Project Deliverables (70%)

The marking criteria for **Project Deliverables** will be based on the following:

- Application 40% (100 marks)
  - Deliverable [1] 10% (25 marks)
  - Deliverable [2] 30% (75 marks)
- Report 20% (100 marks)
- Presentation and Demonstration 10% (100 marks)

Application 40% (100 marks)				
Deliverable [1] 10% (25 marks)				
A (>= 20 marks)	**B (20 > marks >= 17.5)**	**C (17.5 > marks >= 15)**	**D (15 > marks >= 12.5)**	**F (< 12.5 marks)**
User Interface & Interactivity: The web application looks aesthetically pleasing, is highly interactive, intuitive to navigate, and responsive. All links and functionalities work correctly, with no visible bugs or errors. UI/UX design shows creativity and polish beyond classroom examples.	**User Interface & Interactivity:** A working website with some attention to aesthetics and interactivity. Minor usability issues or occasional broken links/errors. Layout is functional but could be improved. Moderate effort in customising beyond lab materials.	**User Interface & Interactivity:** A basic website with limited interactivity and design effort. Several usability issues present. Noticeable broken links or bugs. Appearance lacks polish. Heavy reliance on lab materials, with minimal adaptation.	**User Interface & Interactivity:** Website functions minimally. Lacks aesthetic design and user-friendliness. Many broken links or major bugs. Clear signs of little customisation or incomplete work.	**User Interface & Interactivity:** Website has no aesthetic consideration (e.g., looks like Craigslist). Evidence shows it is copied directly from lab materials with no modification. Non-functional or mostly broken.

Deliverable [2] 30% (75 marks)				
A (>= 60 marks)	**B (60 > marks >= 52.5)**	**C (52.5 > marks >= 45)**	**D (45 > marks >= 37.5)**	**F (< 37.5 marks)**
Cloud Security: Robust data protection practices implemented (e.g., correct bucket policies, HTTPS enforced, and principle of least privilege applied). **Database Functions**: Full and complete implementation of all CRUD operations (Create, Read, Update, Delete) with no errors; properly integrated with front-end and secured. **Deployment of Website:** Website fully deployed on S3 or appropriate cloud platform with seamless functionality, no runtime errors, and responsive across devices. **Rekognition Service:** Rekognition correctly identifies relevant images that align with the website's domain. There is clear interactivity and functional integration with no hardcoded results.	**Cloud Security:** Some appropriate security configurations in place (e.g., non-root MySQL user, basic S3 permissions). **Database Functions:** Most CRUD functions implemented and working, with minor issues. **Deployment of Website:** Successfully deployed, but with occasional errors or minor bugs. **Rekognition Service:** Rekognition runs but may produce irrelevant results, include hardcoded elements, or lack proper user interaction. **Additional Cloud Services:** One or two additional cloud services added that provide some improvement to the website.	**Cloud Security:** Basic security applied (e.g., password authentication used, but no secure S3 policies). **Database Functions:** Only basic Create/Read functions implemented; some errors occur during operation. **Deployment of Website:** Website is deployed but has frequent issues or does not perform consistently. **Rekognition Service:** Rekognition implemented but not meaningfully integrated or partially functional. **Additional Cloud Services:** Minimal or basic use of additional cloud services with limited impact on the overall application.	**Cloud Security:** Weak or insufficient security measures (e.g., default credentials used, no bucket policy). **Database Functions:** Incomplete or faulty CRUD functionality; key operations fail or are missing. **Deployment of Website:** Website deployed but mostly broken or unusable. **Rekognition Service:** Service is not fully working or produces irrelevant/non-functional results. **Additional Cloud Services:** Additional cloud services are poorly implemented or not clearly adding value.	**Cloud Security:** No meaningful data protection or security configuration applied. **Database Functions:** CRUD functions are largely missing or fail to execute. **Deployment of Website:** Website not deployed or non-functional. **Rekognition Service:** Not implemented. **Additional Cloud Services:** No additional services beyond what was demonstrated in lab materials.

Additional Cloud Services: Multiple additional cloud services implemented creatively, showing clear understanding of new knowledge not taught in class, and significantly enhancing the functionality or user experience of the site.				

Report 20% (100 marks)				
A (>= 80 marks)	**B (80 > marks >= 70)**	**C (70 > marks >= 60)**	**D (60 > marks >= 50)**	**F (< 50 marks)**
Development Documentation of AWS Rekognition: Thoroughly documents the development process for integrating and using AWS Rekognition, including custom configurations, handling data inputs and outputs, and optimizing performance. **Deployment Process**	**Development Documentation of AWS Rekognition**: Documents basic usage and integration of AWS Rekognition, but lacks comprehensive details. **Deployment Process Documentation**: Includes the fundamental deployment steps with basic configurations but omits	**Development Documentation of AWS Rekognition:** Describes the use of Rekognition, but with limited explanation of customisation or performance aspects. Data handling and integration are partially described. **Deployment Process Documentation:** Provides a	**Development Documentation of AWS Rekognition:** Limited documentation that briefly mentions Rekognition usage, with missing details on input handling or integration. **Deployment Process Documentation:** Incomplete steps or unclear instructions for deploying the	**Development Documentation of AWS Rekognition**: Documentation on the use of AWS Rekognition is missing or inadequate. **Deployment Process Documentation**: Deployment documentation is incomplete or missing significant elements.

Documentation: Provides an exhaustive, step-by-step guide to deploying the application on AWS, detailing configuration settings, service dependencies, and advanced troubleshooting tips. **Setup Guide for AWS Services**: The setup guide is meticulous and thorough, covering configuration and usage of all AWS services involved, including detailed explanations of the roles and interactions of each service. **Technical Accuracy and Clarity**: The report is technically accurate across all described services and functionalities, using precise and clear language that enhances comprehension. **Visual Aids and Formatting:** Includes effective visual aids (flowcharts, diagrams, screenshots) that significantly enhance the explanation of processes and configurations; the report is excellently formatted for	complex details and advanced troubleshooting. **Setup Guide for AWS Services**: Setup guide covers essential AWS services but lacks depth in explaining interactions or configurations of some secondary services. **Technical Accuracy and Clarity**: Contains inaccuracies that might confuse the understanding; language and technical descriptions are unclear at times. **Visual Aids and Formatting:** Some visual aids are included; they provide basic support to the text; formatting is adequate but could be improved.	general overview of deployment with some configuration details, but lacks clarity on advanced aspects or troubleshooting. **Setup Guide for AWS Services:** Includes setup instructions for main services, but lacks completeness or clarity on service interactions. **Technical Accuracy and Clarity:** Some inaccuracies and vague explanations are present; technical language is inconsistently applied. **Visual Aids and Formatting:** Contains a few visuals, but some may be unclear or irrelevant. Formatting shows effort but may be inconsistent**.**	application. Misses important configuration or dependency details. **Setup Guide for AWS Services:** Basic setup instructions are present but difficult to follow or lacking logic. Interactions between services are not explained. **Technical Accuracy and Clarity:** Frequent technical mistakes or confusing descriptions that make understanding difficult. **Visual Aids and Formatting:** Few or poorly used visuals. Formatting may appear cluttered or inconsistent, reducing readability.	**Setup Guide for AWS Services**: The setup guide is poorly developed, missing, or fails to adequately explain how to configure and use the services. **Technical Accuracy and Clarity:** The technical descriptions are incorrect or so poorly written that they mislead or confuse the reader. **Visual Aids and Formatting:** Lacks visual aids; formatting is poor or non-existent, hindering readability.

easy navigation and readability.				

Project Presentation & Demonstration 10% (100 marks)				
A (>= 80 marks)	**B (80 > marks >= 70)**	**C (70 > marks >= 60)**	**D (60 > marks >= 50)**	**F (< 50 marks)**
Slides Content: All deployment steps were fully documented and well explained; able to accurately describe the function(s) of the services used; able to display deep understanding of AWS services/serverless deployment. **Delivery & Clarity of Presentation**: Delivery of presentation was fluent and well-paced, expressive, and able to explain the topic well. **Presentation Slides Format**: Slides were well prepared; not cluttered, easy to read. **Ability to explain the application**: Able to explain the development of the application confidently and clearly, with little ambiguity.	**Slides Content:** Most deployment steps are clearly documented and explained; able to describe the functions of services used with reasonable accuracy; demonstrates good understanding of AWS/serverless deployment with minor inaccuracies. **Delivery & Clarity of Presentation:** Delivery is smooth, generally well-paced, and expressive; explains the topic effectively with minor gaps or hesitations. **Presentation Slides Format:** Slides are well-organised and mostly easy to read; may have slight clutter or formatting inconsistencies that do not detract significantly.	**Slides Content:** Most of the deployment steps were fully documented and explained; able to describe the function(s) of the services used but some information is not accurate; able to display understanding of AWS services/serverless deployment. **Delivery & Clarity of Presentation**: Delivery of presentation was mostly fluent and appropriately paced, able to explain the topic relatively well. **Presentation Slides Format**: Slides were well prepared but cluttered and hard to read. **Ability to explain the application**: Able to explain the development of the application but not confident	**Slides Content:** Only partial deployment steps documented; explanations are simplistic or lack clarity; understanding of AWS/serverless deployment is superficial. **Delivery & Clarity of Presentation:** Presentation is rushed or lacks structure; explanation of topic is unclear or partially correct. **Presentation Slides Format:** Slides are functional but inconsistently formatted, with readability issues (e.g., small fonts, poor colour contrast). **Ability to Explain the Application:** Attempted to explain the application, but frequently vague or reliant on script/notes.	**Slides Content:** Some of the deployment steps were documented and explained; description of the function(s) of the services used were vague and inaccurate; able to display some understanding of AWS services/serverless deployment. **Delivery & Clarity of Presentation**: Presentation was unprepared; unable to fully explain the topic. **Presentation Slides Format**: Slides were done with slipshod effort; cluttered; wordy. **Ability to explain the application**: Ambiguous / poor explanations on the development of the application. **Ability to Answer Questions**: Able to answer

Ability to Answer Questions: Able to answer all questions confidently and clearly. **Demonstration**: Description of all services used, and their utilisation were demonstrated in detail, utilising the allocated time effectively.	**Ability to Explain the Application:** Clear explanation of the application's development process, though not as in-depth or confident as top-tier presentations. **Ability to Answer Questions:** Able to answer most questions clearly and correctly, though may require clarification or support for more complex ones. **Demonstration:** Services used are demonstrated clearly and logically; makes good use of the allotted time with minor inefficiencies or missing explanations.	and somewhat incoherent, with some ambiguity. **Ability to Answer Questions**: Able to answer most questions confidently and clearly. **Demonstration**: Description of services used, and their utilisation were demonstrated mostly in detail, utilising the allocated time fairly effectively.	**Ability to Answer Questions:** Able to answer only basic questions; shows hesitation or uncertainty in more complex queries. **Demonstration:** Attempted to demonstrate the application but lacked depth or clarity in showing how services were used. Time was poorly managed.	questions confidently and clearly with some doubts **Demonstration**: Description of services used, and their utilisation were demonstrated briefly, not utilising the allocated time effectively.

END OF PROJECT SPECIFICATIONS